

Lightweight Approach for Multi-Modal Irony Detection by Image Caption Generation

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Abstract—In this work, we tackle the multi-modal irony detection problem and propose a lightweight approach that combines pretrained multi-modal and text-based smaller language models. The proposed approach makes use of generated image captions and original text content to fine-tune a transformer based model for text classification. In the proposed solution, Vision-Language Model (VLM) is used for generating image captions, without any fine-tuning. On two benchmark datasets, the experiments demonstrate the benefit of the proposed method against the baseline solutions.

Index Terms—Multimodal Irony Detection, Sarcasm Detection, Large Language Models, Vision-Language Models, Social Media

I. INTRODUCTION

Irony is a form of figurative speech that is basically characterized with the difference between what is said and what is really meant. It is often used to criticize or make fun of a situation, particularly when something negative or undesired happens. While irony traditionally refers to situational contradictions and sarcasm involves intentional mockery, both phenomena share the core characteristic of meaning inversion. Therefore, in this work, we treat irony and sarcasm detection as a unified computational problem. On social media, irony and sarcasm are common rhetorical devices, making their detection an important challenge in Natural Language Processing (NLP). Earlier approaches rely solely on text [1] or images [2], but such single-modality methods often miss contextual cues. For instance, the post “Great, another Monday morning” may appear neutral in text, but becomes clearly ironic when paired with an image of frustration. Moreover, as shown in Figure 1, the posts and the images might contain contrast information that are necessary to notice the sarcasm. These phenomenon motivates multimodal approaches that jointly model textual and visual information [2].

Multimodal irony detection typically follows two strategies. Some studies employ unified Vision-Language Models (VLMs) that process both text and image together, learning joint representations. For example, Cai et al. [3] propose a transformer-based framework that receives both modalities simultaneously. Others adopt fusion-based designs, where each

modality is encoded separately and later combined. For instance, Qin et al. [4] use a CNN to extract visual features and an LSTM for textual features, which are then merged via a fusion layer. Both strategies outperform unimodal baselines, though unified VLMs excel at cross-modal relations, while fusion-based systems remain attractive for their efficiency.

In this work, we propose a simple yet effective approach: images are first converted into textual descriptions using VLMs (Qwen-VL 2.5–7B, InternVL3-8B). These captions are concatenated with the original post text and then classified using a fine-tuned transformer-based text models (RoBERTa, ModernBERT). This design reduces multimodal processing to a text-only input format, enabling resource-friendly training without full multimodal finetuning. Our main contributions are as follows:

- We formulate multimodal irony detection as a text classification task by representing image content as natural-language descriptions, enabling the use of strong text encoders without multimodal finetuning.
- The proposed approach is evaluated on two widely adopted datasets: MMSD2.0 [4] (24k ironic and non-ironic posts) and MORE [5]. We examine the benefit of leveraging text and image information on these datasets.
- We empirically demonstrate that our lightweight image caption based approach outperforms multi-modal zero-shot baselines, and provides competitive or superior results compared to state-of-the-art multimodal baselines such as Multi-view CLIP.

II. RELATED WORK

A. Text-only Irony Detection

Earlier research on irony and sarcasm detection primarily relied on traditional machine learning approaches with hand-crafted features. Studies employed Support Vector Machines and Naive Bayes (NB) classifiers with linguistic features such as n-grams, sentiment polarity, and syntactic patterns [5]–[7]. Later work also investigated satire on social media; for instance, Barbieri et al. [8] analyzed satirical and sarcastic expressions in Twitter posts. Subsequently, neural network architectures, particularly LSTM-based models, were explored for capturing sequential patterns in sarcastic text [9]–[11].

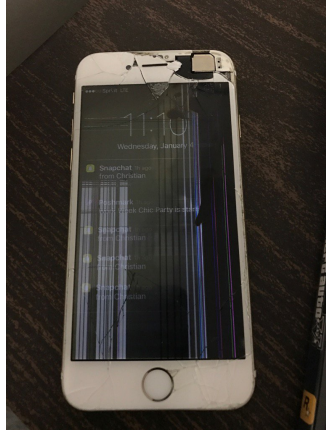
When the holidays are over and you gotta face reality again



MMSD2.0 Example

Original Text:

“Why hello reality, nice of you to creep back so fast.”



MORE Example

Original Text:

“My sister treats her lease very well.”

Fig. 1. Examples of ironic multimodal posts from MMSD2.0 and MORE.

The introduction of transformer architectures marked a significant advancement in the field. BERT-based models [12]–[16] achieved substantial improvements across irony detection benchmarks, with studies focusing on fine-tuning pre-trained language models for sarcasm detection. Savini and Caragea [17] further showed that intermediate-task transfer learning with BERT yields strong baselines for sarcasm detection. Furthermore, recent work has explored prompt engineering techniques with large language models [1], [18]–[20], demonstrating competitive zero-shot and few-shot performance for irony detection tasks.

B. Multimodal Irony Detection

While text-only approaches dominated early research, the recognition that irony often depends on visual context led to the development of multimodal datasets and methods. Schifanella et al. [21] pioneered multimodal sarcasm detection by introducing datasets pairing text with images from social media, demonstrating that visual cues significantly improve detection performance. Goyal et al. [22] introduced a multimodal dataset extending MUSTARD with satire and sarcasm annotations and proposed a BERT-based model for classification.

Recent methodologies have leveraged transformer architectures for multimodal fusion. Vision-language transformers have been adapted for sarcasm detection using disentangled text attention mechanisms and visual transformers [23], while cross-modal attention mechanisms capture incongruity between textual and visual content [24], [25]. Recently, Nandy et al. [2] introduced the YesBut dataset for satirical image understanding, providing tasks such as detection, explanation, and completion, and showed that state-of-the-art vision–language models perform poorly in zero-shot settings. Unified multimodal encoders like VisualBERT [26] and contrastive models

such as CLIP [27] have achieved strong performance on datasets including MMSD2.0 [4]. Liu et al. [28] further addressed the inherent uncertainty in multimodal sarcasm and sentiment detection by introducing a quantum probabilistic framework, achieving state-of-the-art results.

C. The Proposed Approach vs SoTA

Our approach differs from these prior works by adopting a caption-then-classify pipeline that converts images into natural-language descriptions using state-of-the-art VLMs (Qwen-VL 2.5–7B, InternVL3-8B). These captions are concatenated with original text and processed using lightweight transformer-based classifiers (RoBERTa, ModernBERT). This strategy enables the use of powerful pre-trained text encoders without requiring multimodal fine-tuning, offers a resource-friendly alternative to end-to-end multimodal training, and leverages strong captioning capabilities.

III. METHOD

We propose a multimodal irony detection pipeline that integrates visual and textual signals through pretrained vision–language and transformer models. The approach consists of three basic components: image description generation, merging the information from text and image modalities, and training a classifier for sarcasm detection, as shown in Figure 2. If the data includes hashtags and emojis, additionally, a data preparation and processing step is also applied.

A. Image Caption Generation

To incorporate visual context, image component of the data is converted into a natural-language caption using a vision–language model. While building the solution, we used Qwen-VL 2.5–7B, InternVL3-8B models. These models are selected on the basis of the performance results reported in the benchmarks such as Open VLM Leaderboard¹, and initial experiments to observe the caption generation performance of VLMs. These selected models are applied in inference mode with the following prompt:

“Please describe this image in detail. Focus on the visual elements, composition, and any notable features. Keep the description under 250 tokens.”

The generated captions provide detailed descriptions that summarize the visual modality and are treated as auxiliary text in the next phase.

B. Merging Text and Image Information

We concatenate the original post with the generated caption, creating a unified textual input, as shown in Figure 3. This formulation translates the multimodal irony detection problem into a purely textual classification task, enabling the direct use of pretrained text-only language models.

¹https://huggingface.co/spaces/opencompass/open_vlm_leaderboard

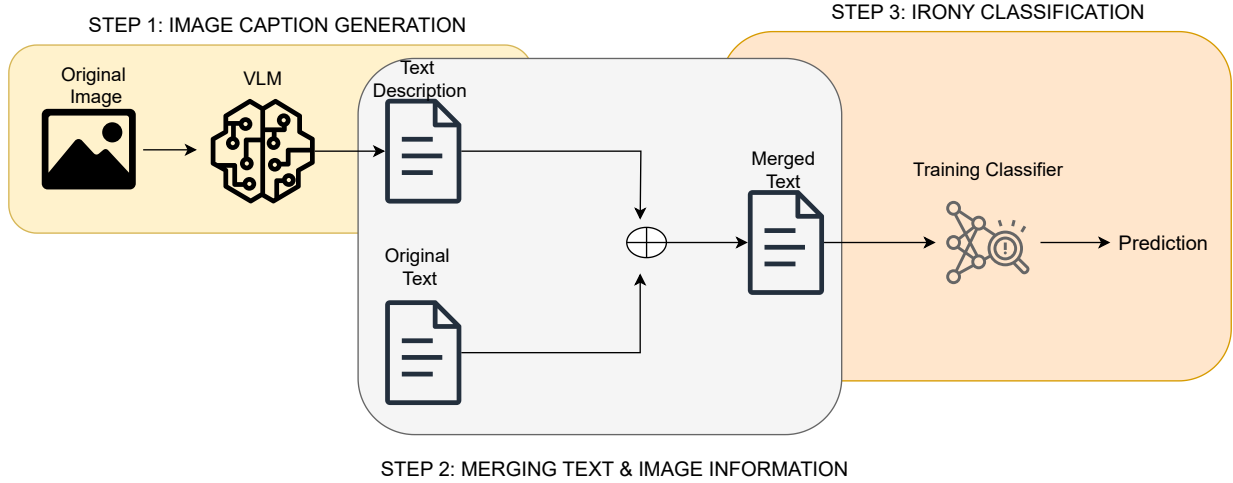


Fig. 2. Overview of the proposed multimodal irony detection pipeline.



(a) Image.

Original Text: “feeding my abs nothing but the best quality beef”

Generated Caption: “The image shows a meal consisting of two burgers and a serving of fries. The burgers are placed in individual cardboard containers with sesame seed buns, visible layers of meat, lettuce, and possibly cheese. The fries are served in a paper container with a striped design, sitting on a textured surface that appears to be a table or countertop. The overall setting suggests a casual dining environment, likely from a fast-food restaurant.”

Combined Input: “feeding my abs nothing but the best quality beef The image shows a meal consisting of two burgers and a serving of fries. The burgers are placed in individual cardboard containers with sesame seed buns, visible layers of meat, lettuce, and possibly cheese. The fries are served in a paper container with a striped design, sitting on a textured surface that appears to be a table or countertop. The overall setting suggests a casual dining environment, likely from a fast-food restaurant.”

(b) Textual representation.

Fig. 3. Example post demonstrating multimodal input construction with (a) the original image and (b) its textual representation (original text, generated caption, and combined input).

C. Irony Classification

At this point, the multimodal data instances are re-represented in text-only modality. In this way, we aim to exploit the capability of transformer-based smaller language models to process and analyze textual data. The combined text includes the original text component of the data and the caption of the image component. By this way, we can convert

a multi-modal irony dataset to a text-only collection, in order to train a classifier with textual input. In our proposed solution, we use transformer-based classifiers, particularly RoBERTa and ModernBERT. We selected these models on the basis of performance results in the literature and the initial experiments conducted.

Once the model is trained by using the text-only collection, in order to classify a given data instance, we first obtain a text-only representation of it (including the original text and the generated caption). The combined text is tokenized and passed to transformer-based classifiers (RoBERTa, ModernBERT) for binary irony detection. This design provides a lightweight yet effective alternative to end-to-end multimodal architectures, leveraging visual cues while maintaining the efficiency of text-only training.

D. Dataset Preparation Phase

We evaluate our approach on two multimodal sarcasm datasets: MMSD2.0 [4] and MORE [29]. MMSD2.0 consists of multimodal posts collected from Twitter, Instagram, and Tumblr, pairing user text with associated images. The dataset was initially gathered via *distant supervision*, using hashtags such as #sarcasm and #irony, and subsequently refined through **manual annotation** to remove false positives and balance sarcastic vs. non-sarcastic classes. The final corpus contains 24,635 posts, split into training (19,816), validation (2,410), and test (2,409). Class distributions are balanced across splits: the training set includes 9,572 sarcastic vs. 10,240 non-sarcastic posts, while the test set includes 1,037 sarcastic vs. 1,372 non-sarcastic posts.

The MORE dataset was released for multi-modal sarcasm explanation rather than sarcasm detection, combining sarcastic posts from multiple existing datasets. Unlike MMSD2.0, MORE focuses on generating natural language explanations for the underlying irony in sarcastic content. In other words, there does not exist any non-sarcastic instance in the MORE

dataset. The dataset contains 3,510 sarcastic posts with corresponding explanations, split into training (2,983), validation (175), and test (352). All samples are sarcastic and paired with manually generated explanations created by expert annotators following comprehensive annotation guidelines to ensure explanation quality and consistency.

To ensure consistency across datasets, we apply the same cleaning strategy used in MMSD2.0 to MORE. Specifically, we remove emojis and spurious hashtags while preserving the semantic text. Since MORE is designed for explanation generation while MMSD2.0 focuses on sarcasm detection, we adapt MORE for our detection task by using the presence of explanations as positive sarcasm indicators. We then merge the cleaned MORE dataset with MMSD2.0, yielding a larger and more balanced resource. This augmentation increases the training size by roughly 3.5k sarcastic posts, improving robustness to noise and domain variation.

IV. EXPERIMENTS

A. Experimental Setup

We evaluate our approach on MMSD2.0 [4] and MORE [29] datasets. MORE dataset is used as an additional source in a merged setup following the preprocessing steps described in Section III-D. In the experiments, the partition ratios for training, validation and test are 80%, 10% and 10%.

All models are trained for up to 5 epochs using AdamW optimizer with a cosine learning rate schedule. The best checkpoint is selected by validation macro-F1 score. Training is conducted on GPUs with ≥ 40 GB memory. We report accuracy, macro-F1, precision, and recall scores, with macro-F1 as the primary metric. Note that all experiments are conducted on the MMSD2.0 test set unless stated otherwise.

For the zero-shot baseline experiments, we employ zero-shot prompting with Qwen-VL 2.5-7B [30] and Mistral-7b [31]. Original posts with corresponding images are directly given as prompts to VLMs. The prompts used in the zero-shot baselines are given in Appendix A.

For the supervised fine-tuning evaluation, we fine-tune RoBERTa (base/large) and ModernBERT (base/large) on concatenated text+caption inputs with cross-entropy loss. We search over learning rates $\{2e-5, 5e-5\}$ and batch sizes $\{4, 8, 16\}$, selecting the best configurations by validation F1 score. The results of these hyperparameter optimization experiments are given in Appendix B.

The proposed pipeline is compared against the Multi-view CLIP baseline [4] that utilizes text, image and text-image interaction views. We reproduced the model as explained in the paper and trained the model according to the presented setting.

B. Irony Detection Performance Experiments

Table I presents a unified comparison across zero-shot baselines, single modality baselines, Multi-view CLIP baseline, and our caption-then-classify model.

In the analysis, zero-shot evaluation provides a natural baseline: the *vanilla prompt* (in Appendix A) on Qwen-VL

2.5-7B achieves the best result (67.73% F₁), outperforming both the detailed prompt (60.99% F₁) and the Mistral-7B setup (61.05% F₁). Although the results are significantly lower than the supervised fine-tuned models, these results indicate the raw performance of the selected VLMs on the multimodal irony detection problem.

Concatenating generated captions to the original text (text+caption) improves over the text-only baseline confirming that visual information adds complementary cues. InternVL and Qwen-VL 2.5-7B produce near-identical results once modalities are concatenated, indicating that caption quality differences are largely neutralised by the fusion step.

In comparison Multi-view CLIP, the our proposed solution provides higher score for Precision, same score for F1 and very close scores for accuracy and recall, by applying fine-tuning only for text modality.

C. Data Augmentation Analysis

In order to analyze the effect of including additional data, we integrated the full MORE dataset while preserving its original train-validation-test split partitions. MORE dataset is preprocessed as described in Section III-D and then combined with MMSD2.0. We also re-implemented and ran the official Multi-View CLIP code in our environment on this merged benchmark to ensure a fair comparison under the same data splits. Adding the cleaned MORE corpus (~ 3.5 k posts) further lifts performance. As shown in Table II, ModernBERT gains **+1.14% F1**, surpassing Multi-View CLIP trained and evaluated on the same merged data (85.95% vs. 84.81% F1). The extra data mainly reduces sarcastic false negatives (20% drop), demonstrating that the proposed pipeline particularly benefit from larger and more diverse training sets.

D. Text Classifier Analysis

In Table III, the results for our caption-then-classify model with different text classifiers are presented. Additionally, we performance with the use of InternV3-8B is also given in comparison to Qwen2.5-VL-7B. Although InternV3 model provides better detection scores in single modalities (see Table I), for the image caption + text setting, its performance falls behind Qwen2.5-VL-7B. Hence, for the performance analysis of text classifiers, we only used Qwen2.5-VL-7B for caption generation.

In the text classifier comparisons, replacing XLM-RoBERTa with ModernBERT yields the strongest supervised result (84.59% F1), nearly matching Multi-view CLIP (84.64% F1) without any end-to-end multimodal fine-tuning. This suggests that architectural advances in text backbones can offset the lack of joint visual training.

E. Discussion

Our experimental results demonstrate the effectiveness of multimodal approaches for irony detection. As shown in Table I, incorporating visual information through caption concatenation consistently improves performance across all model configurations. The text+caption combination achieves 82.00%

TABLE I
COMPARISON OF THE PROPOSED METHOD WITH THE BASELINES UNDER ZERO-SHOT AND FINE-TUNED SETTINGS ON MMSD2.0.

Model	Input	Image Encoder	Text Encoder	Inf. Setup	Acc (%)	F1 (%)	Prec (%)	Rec. (%)
Zero-shot-1	image + text – vanilla	Dynamic ViT	Qwen2.5 LLM	Zero-shot	69.12	67.73	66.78	56.22
Zero-shot-2	image + text – detailed	Dynamic ViT	Qwen2.5 LLM	Zero-shot	61.15	60.99	78.40	53.31
Zero-shot-3	image caption + text	–	Mistral-7B	Zero-shot	67.04	61.05	71.52	63.17
Zero-shot-4	text only	–	Mistral AI	Zero-shot	64.80	64.45	64.45	64.68
Zero-shot-5	image caption-only	–	Mistral AI	Zero-shot	64.51	55.87	71.02	59.51
Baseline-text	text-only	Qwen2.5-VL-7B	xlrm-roberta-base	Fine-tuned	78.79	78.58	79.21	78.79
Baseline-text	text-only	InternVL3-8B	xlrm-roberta-base	Fine-tuned	78.33	78.15	78.87	78.33
Baseline-caption	image caption-only	Qwen2.5-VL-7B	xlrm-roberta-base	Fine-tuned	74.55	74.19	74.74	74.55
Baseline-caption	image caption-only	InternVL3-8B	xlrm-roberta-base	Fine-tuned	75.34	75.02	75.59	75.34
MV CLIP [4]	image + text	clip-vit-base	CLIP Transformer	Fine-tuned	84.81	84.64	84.49	86.42
Ours	image caption + text	Qwen2.5-VL-7B	ModernBERT-large	Fine-tuned	84.72	84.59	85.21	84.72

TABLE II
EFFECT OF AUGMENTING MMSD2.0 WITH CLEANED MORE DATA.

Model	Input	Image Encoder	Text Classifier	Inf. Setup	Acc (%)	F1 (%)	Prec (%)	Rec. (%)
MV CLIP [4]	image + text	clip-vit-base	CLIP Transformer	Fine-tuned	84.97	84.81	84.66	86.60
Ours	image caption + text	Qwen2.5-VL-7B	xlrm-roberta-large	Fine-tuned	84.28	84.27	84.34	84.28
Ours	image caption + text	Qwen2.5-VL-7B	ModernBERT-large	Fine-tuned	85.98	85.95	86.27	85.98

TABLE III
TEXT CLASSIFIER ANALYSIS RESULTS

Input	Image Encoder	Text Classifier	Acc (%)	F1 (%)	Prec (%)	Rec. (%)
image caption + text	Qwen2.5-VL-7B	xlrm-roberta-base	82.27	82.00	82.39	82.27
image caption + text	InternVL3-8B	xlrm-roberta-base	82.57	82.42	83.10	82.57
image caption + text	Qwen2.5-VL-7B	xlrm-roberta-large	83.48	83.35	84.04	83.48
image caption + text	Qwen2.5-VL-7B	ModernBERT-large	84.72	84.59	85.21	84.72

F1 (in Table III) with RoBERTa-base classifier, representing a substantial improvement over the text-only baseline (78.58% F1 in Table I). This confirms that visual context provides complementary cues that are essential for understanding ironic intent in social media posts.

Zero-shot evaluation with vision-language models reveals interesting but limited capabilities. While Qwen-VL 2.5-7B with simple prompting achieves 67.73% F1, this performance falls significantly short of supervised approaches. The gap between zero-shot (67.73% F1) and our best supervised model (84.59% F1 with ModernBERT) highlights the necessity of task-specific training. Rather than pursuing computationally costly end-to-end multimodal fine-tuning, our lightweight pipeline leverages pre-trained vision-language models for caption generation and focuses computational resources on training efficient text encoders. This strategy proves highly effective, with ModernBERT-large achieving 84.59% F1, nearly matching the multi-view CLIP baseline (84.64% F1) without requiring joint visual-textual training.

The robustness of our approach becomes particularly evident when examining cross-domain generalization. While our model initially performs slightly below multi-view CLIP on the original MMSD2.0 test set, the gap narrows considerably when both models are trained on the merged MMSD2.0+MORE dataset. Our caption-based

pipeline achieves 85.95% F1 compared to multi-view CLIP’s 84.81% F1 on the same augmented data, suggesting superior adaptability to diverse domains.

V. CONCLUSION

We introduced a lightweight multimodal irony/sarcasm detection pipeline that first translates images into natural language descriptions and then feeds the concatenated text+caption content to an irony classifier. This “caption-then-classify” design avoids end-to-end multimodal finetuning while still exploiting visual cues. As a future work, we plan to extend our experiments to more diverse datasets (including other platforms and languages). Beyond open-source models, we also aim to evaluate proprietary multimodal systems such as Gemini for comparative analysis.

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APPENDIX A ZERO-SHOT PROMPTS

Qwen-VL 2.5-7B — Vanilla Prompt

You are analyzing a post containing both text and an image. Determine if the post conveys irony or sarcasm. If ironic/sarcastic → respond with exactly '1'. If not ironic → respond with exactly '0'. Consider both text and image. Respond only with 0 or 1.

Qwen-VL 2.5-7B — Detailed Prompt

You are an expert in multimodal sarcasm and irony detection. Key indicators: (1) Contrast between text and image (2) Hyperbolic expressions (3) Situational irony. Response: '1' if sarcastic/ironic, '0' if literal. Only output 0 or 1.

Mistral AI (open-mistral-7b)

Determine if the following text is ironic (1) or not ironic (0). Irony is when words express something different from or opposite to their literal meaning, often for humorous effect. Answer only with 0 (not ironic) or 1 (ironic).

APPENDIX B HYPERPARAMETER VALIDATION EXPERIMENTS

TABLE IV

QWEN-VL 2.5-7B + ROBERTA-BASE (TEXT-ONLY) ON MMSD2.0 TEST.

Config (lr, bs)	F1	Acc	Prec	Rec
(5e-05, 4)	66.37	66.67	67.21	66.67
(2e-05, 4)	78.58	78.79	79.21	78.79
(5e-05, 8)	76.58	76.67	77.79	76.67
(2e-05, 8)	77.41	77.63	78.07	77.63
(5e-05, 16)	77.42	77.54	78.40	77.54
(2e-05, 16)	78.09	78.25	78.90	78.25

TABLE V

QWEN-VL 2.5-7B + ROBERTA-BASE (IMAGE CAPTION-ONLY) ON MMSD2.0 TEST.

Config (lr, bs)	F1	Acc	Prec	Rec
(5e-05, 4)	68.03	68.83	68.69	68.83
(2e-05, 4)	73.60	73.97	74.15	73.97
(5e-05, 8)	36.29	56.95	32.44	56.95
(2e-05, 8)	72.82	73.18	73.41	73.18
(5e-05, 16)	74.03	74.39	74.59	74.39
(2e-05, 16)	74.19	74.55	74.74	74.55

TABLE VI

INTERNVL3-8B + ROBERTA-BASE (TEXT-ONLY) ON MMSD2.0 TEST.

Config (lr, bs)	F1	Acc	Prec	Rec
(5e-05, 4)	36.29	56.95	32.44	56.95
(2e-05, 4)	69.29	69.49	70.24	69.49
(5e-05, 8)	65.28	65.34	66.89	65.34
(2e-05, 8)	77.45	77.67	78.12	77.67
(5e-05, 16)	78.15	78.33	78.87	78.33
(2e-05, 16)	77.15	77.33	77.89	77.33

TABLE VII

INTERNVL3-8B + ROBERTA-BASE (IMAGE CAPTION-ONLY) ON MMSD2.0 TEST.

Config (lr, bs)	F1	Acc	Prec	Rec
(5e-05, 4)	67.81	68.12	68.60	68.12
(2e-05, 4)	73.81	74.01	74.64	74.01
(5e-05, 8)	75.02	75.34	75.59	75.34
(2e-05, 8)	74.02	74.30	74.68	74.30
(5e-05, 16)	73.28	73.56	73.96	73.56
(2e-05, 16)	74.77	75.01	75.49	75.01

TABLE VIII

QWEN-VL 2.5-7B + MODERNBERT-LARGE (TEXT-ONLY) ON MMSD2.0 TEST.

Config (lr, bs)	F1	Acc	Prec	Rec
(1e-05, 8)	83.87	84.02	84.46	84.02
(2e-05, 8)	84.59	84.72	85.21	84.72
(5e-05, 8)	81.58	81.74	82.24	81.73
(1e-05, 16)	82.84	82.94	83.81	82.94
(2e-05, 16)	83.09	83.19	84.06	83.19
(5e-05, 16)	81.14	81.19	82.54	81.20

TABLE IX

QWEN-VL 2.5-7B + ROBERTA-LARGE ON MMSD2.0+MORE (CLEANED).

Config (lr, bs)	F1	Acc	Prec	Rec
(2e-05, 16)	84.27	84.28	84.34	84.28
(1e-05, 8)	83.85	83.85	83.85	83.85
(1e-05, 16)	83.02	83.05	83.23	83.05

TABLE X

QWEN-VL 2.5-7B + MODERNBERT-LARGE ON MMSD2.0+MORE (CLEANED).

Config (lr, bs)	F1	Acc	Prec	Rec
(1e-05, 16)	84.56	84.57	84.69	84.57
(2e-05, 16)	83.71	83.74	83.87	83.74
(5e-05, 16)	85.95	85.98	86.27	85.98